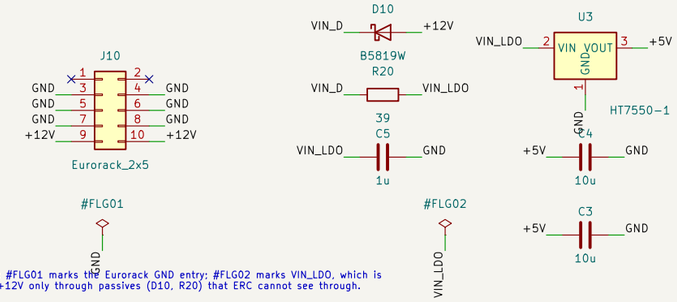
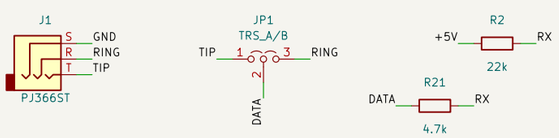
BIG MODULAR MIDI — S3 schematic (5 V logic). Circuit intent: docs/research.md

POWER: Eurorack +12V -> B5819W -> 39R -> HT7550-1 -> +5V. -12V (J10 pins 1-2) unused.



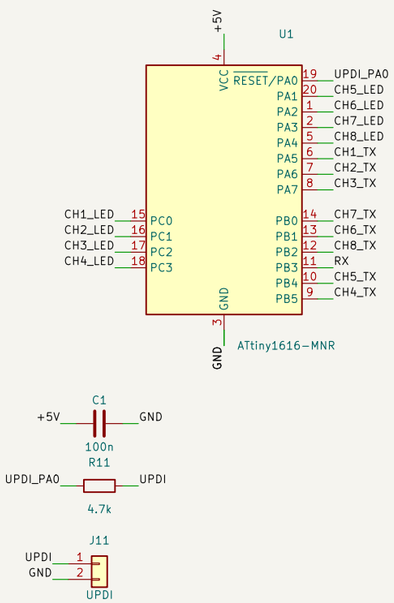
PWR_FLAG: #FLG01 marks the Eurorack GND entry; #FLG02 marks VIN_LDO, which is fed from +12V only through passives (D10, R20) that ERC cannot see through.

MIDI IN: Bread Modular V1 direct input (3.3 V and 5 V senders), no opto



JP1 TRS TYPE (2.54 mm header on the back): shunt 1-2 = TYPE A (Tip = data), shunt 2-3 = TYPE B (Ring = data). The unused conductor is left open. Sleeve = GND (as Bread Modular V1). Firmware: no PORT_INVEN. Opto-isolated input (ground-loop break, voltage-sensed) is deferred to V2.

MCU: ATtiny1616 at 5 V, 20 MHz; pin map identical to Bread Modular



UPDI J11: pin 1 UPDI, pin 2 GND (2.54 mm). Power the module from Eurorack while programming.

OUTPUTS 1-8: GPIO -> 1k -> TS tip (sleeve GND); GPIO -> 1k -> LED -> GND



Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S3 — 5 V logic — not released, not for fabrication

Big Modular

Sheet: /
File: midi.kicad_sch

Title: Big Modular — MIDI

Size: A3	Date:	**Rev: S3**
KiCad E.D.A. 10.0.6		Id: 1/1